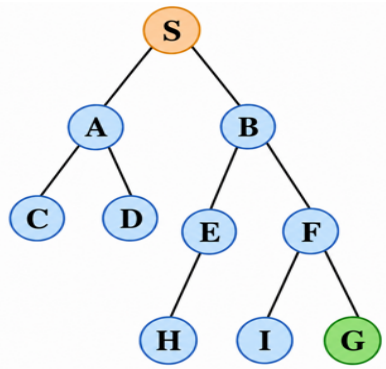
V SEMESTER
Code: 23ADS1502 Sub. Name: Lab: Artificial Intelligence

Course Outcomes
Upon successful completion of the course the students will be able to **CO1:** Develop an understanding to identify appropriate performance measures and environment models for intelligent agents. **CO2:** Apply various search algorithms for goal-based problem solving and planning in deterministic and non-deterministic environments. **CO3:** Construct appropriate knowledge representation models for logical inference. **CO4:** Analyze and solve problems under uncertainty using probabilistic models and decision-theoretic reasoning.

SN	**Experiments based on**
1	**Practical 1:** Smart Light Agent (Simple Reflex Agent) **Problem Statement:** Design and implement a simple reflex agent to simulate a smart light system for a room. The agent decides whether to turn the indoor light ON or OFF based on two percepts: • Outside light (sunlight available or not) • Human presence (someone present in the room or not) The light should behave as follows: • If human is present and outside light is not available, the agent should turn ON the light. • In all other cases, the light should be turned OFF.
2	**Practical 2: Hospital Patient Monitoring Agent** **Problem Statement:** Implement a Model-Based Reflex Agent for monitoring patients in a hospital ward. There are three patients. Each patient may have either Normal or Critical health status. The agent maintains an internal record of every patient's condition. Rules: • If a patient's condition is Critical, alert the doctor. • If the condition is Normal, continue monitoring. • Update the patient's status after every inspection. • Stop after all patients have been checked.

3	**Practical 3**: Shortest Path Using Greedy Best-First Search **Problem Statement:**An AI agent must reach the goal node of a graph as fast as possible. The agent uses a Greedy Best-First Search strategy — always choosing the node that looks closest to the goal.  node | H (n) A | 12 B | 4 C | 7 D | 3 E | 2 H | 4 I | 9 S | 13 G | 0
4	**Practical 4**: Optimal Path Using A Search* **Problem Statement:** A robot moves in a city map where each move has a cost (e.g., traffic or terrain). The goal is to find the **lowest cost path** to the destination using *A search**, combining actual cost and heuristic (distance). **Map Example (Grid + Weights):** [1, 1, 5] [1, X, 1] [1, 1, 1]
5	**Practical 5**: Map Coloring for Adjoining Regions using Min-Conflicts **Problem Statement:** You are given a simple map of 5 regions (A, B, C, D, E), with a few of them sharing borders. Color each region using only 3 colors such that no neighboring regions share the same color. AI Concepts: • CSP (variables: regions, domains: 3 colors) • Local search (min-conflicts algorithm)

6	**Practical 6:** Two-Player Game using Minimax and Alpha-Beta Pruning **Problem Statement:** Develop an AI agent for a two-player game such as **Tic-Tac-Toe**. The AI should: • Generate possible moves. • Construct the game tree. • Apply the Minimax algorithm. • Select the optimal move. • Implement Alpha-Beta pruning to reduce unnecessary node evaluation. • Compare the number of nodes examined with and without pruning.
7	**Practical 7:** Smart Garden System **Problem Statement:** In a smart home garden in India, we have these rules: • If it is sunny, then plants are dry. • If plants are dry, then water is needed. • Today, it is sunny in Nagpur. Use a truth table to determine whether the system will decide to water the plants. Goal: Apply propositional logic to infer conclusions from weather-based rules.
8	**Practical 8:** Assistant Bot using Unification (First-Order Logic – Unification) **Problem Statement:** A chatbot is trained to understand preferences. It has two rules: • Loves(Neha, x) • Loves(y, PaniPuri) Use unification to check whether Neha loves PaniPuri, and if so, what variable substitutions are needed. Goal: Apply unification algorithm to match predicate logic statements.
9	**Practical 9:** Nagpur Metro Route Planner Topic: State-Space Search Planning (Forward Search) Scenario: A metro planner assistant helps Ankit find the shortest path from Sitabuldi to MIHAN using Nagpur Metro. Stations & connections: • Sitabuldi → Airport (10 min) • Airport → New Khapri (8 min) • New Khapri → MIHAN (7 min) Design a forward state-space search planner that finds the optimal route for Ankit from Sitabuldi to MIHAN with minimum travel time.

10	**Practical 10:** Kitchen Task Planner for Anaya Partial Order Planning Anaya wants to make chai (tea). Steps include: • Boil water • Add tea leaves • Add milk • Add sugar • Serve tea Constraints: • Boil water before adding tea leaves • Add milk only after tea leaves are in • Sugar can be added any time after milk Model the partial order plan for making tea efficiently without violating task constraints.